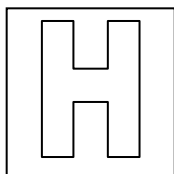


Name:	Index No.:	CT Group: 10
--------------	-------------------	---------------------



PIONEER JUNIOR COLLEGE

2011 JC2 PRELIMINARY EXAMINATION
HIGHER 1



CHEMISTRY

8872/02

Paper 2

19 September 2011

2 hours

Candidates answer Section A on the Question Paper

Additional Materials: Data Booklet
 Answer Paper
 Graph Paper

READ THESE INSTRUCTIONS FIRST

Write your name, CT group and index number on all the work you hand in.

Write in dark blue or black pen.

You may use a pencil for any diagrams, graphs or rough working.

Section A

Answer **all** questions.

Section B

Answer any **two** questions on separate answer paper.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE			
Section A		Section B	
1	/ 15	5	/ 20
2	/ 10	6	/ 20
3	/ 8	7	/ 20
4	/ 7	Penalty	s.f. / units
		TOTAL	/ 80

Section A (40 marks)

Answer all questions in the spaces provided.

- 1 (a) Nitric acid is a highly corrosive liquid and a strong oxidising agent. It can oxidise $\text{Fe}^{2+}(\text{aq})$ to $\text{Fe}^{3+}(\text{aq})$.

In an experiment, 25.0 cm^3 of $0.600 \text{ mol dm}^{-3} \text{ Fe}^{2+}(\text{aq})$ required 25.0 cm^3 of $0.200 \text{ mol dm}^{-3} \text{ HNO}_3$ for complete reaction.

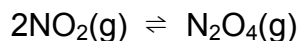
- (i) Complete the table below which refers to possible reduction products of nitric acid.

Formula of product	Oxidation number of nitrogen
NO_2	
NO	
N_2O	
NH_4^+	

- (ii) How many moles of Fe^{2+} react with one mole of HNO_3 ?
- (iii) What change in oxidation number does the nitrogen in HNO_3 undergo?
- (iv) Identify the nitrogen-containing product of this reaction from the table in (i).
- (iv) Construct an equation for the reaction of HNO_3 with Fe^{2+} .

[6]

- (b) In the gaseous state, NO_2 , which is used in the preparation of nitric acid, can dimerise as given in the equation.



- (i) Write an expression for the equilibrium constant, K_c , for the dimerisation of NO_2 .
- (ii) When 1 mol of NO_2 is placed in a 2 dm^3 vessel at room temperature, it is found that the equilibrium mixture contains 0.35 mol of N_2O_4 . Calculate a value of K_c for the reaction, giving its units.
- (iii) The NO_2 molecule has an unpaired electron on the nitrogen atom but the N_2O_4 molecule does not. Using the information, suggest a structural formula for N_2O_4 . Hence, explain whether the dimerisation of NO_2 is endothermic or exothermic.
- (iv) Explain, with reasons, whether the dimerisation above is favoured by
(I) high or low pressure

(II) high or low temperature

[9]

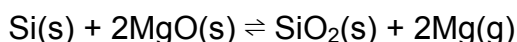
[Total: 15]

- 2 Magnesium is a member of the alkaline earth metals. It occurs in nature, only in the combined state, as *dolomite*, $\text{CaCO}_3\text{MgCO}_3$, *kieserite*, MgSO_4 , *carnallite*, $\text{KCl}\cdot\text{MgCl}_2\cdot 6\text{H}_2\text{O}$, *Epsom salt*, $\text{MgSO}_4\cdot 7\text{H}_2\text{O}$ and many silicates including talc and asbestos.

Magnesium compounds are also present in sea water. Being the second most abundant cation in sea water, it is estimated that magnesium is present in concentrations as high as 1290 parts per million (ppm) and that makes it an attractive commercial source of magnesium. [1 ppm of magnesium = 1 mg of magnesium in 1 dm³ of solution]

About 70% of the Earth's surface is covered by water. The oceans of the planet have an area of about $360 \times 10^6 \text{ km}^2$ with an average depth of about 3.7 km.

As of 2005, China has taken over United States as the dominant supplier of magnesium, pegged at 60% of the world's market share. Unlike the typical use of electrolysis to extract magnesium, China makes use of silicothermic reduction, known as Pidgeon Process. Calcium hydroxide is added to sea water to precipitate magnesium hydroxide. The magnesium hydroxide is then calcined to give magnesium oxide by removal of water. The magnesium oxide and silicon is then heated strongly according to the equation below.



This reaction is not thermodynamically favourable. However, applying Le Chatelier's principle, the position of equilibrium can be shifted to the right by continuous supply of heat and distilling out the magnesium vapour formed. The magnesium extracted through this process is known to have very high purity.

Substance	Melting Point / °C	Boiling Point / °C
Mg	650	1091
MgO	3125	3873
Si	1414	3265
SiO ₂	1700	2230

One of the main applications of magnesium is in the manufacturing of alloys such as duralumin and magnalium. These alloys are used for the construction of aircrafts and racing cars. Magnesium is also used in flares and fire-works.

- (a) Use the data above to estimate the volume of the Earth's oceans in dm³. [2]

- (b) Calculate the concentration, in g dm⁻³, of magnesium in seawater. [1]

(c) Estimate the total mass of magnesium in the ocean. [1]

(d) Account for the high melting and boiling point of magnesium in terms of its structure and bonding. [2]

(e) (i) Suggest a minimum and maximum temperature for the Pidgeon process.

(ii) Explain why magnesium obtained through this process has very high purity.

[2]

(f) Suggest a property of magnesium of its use in

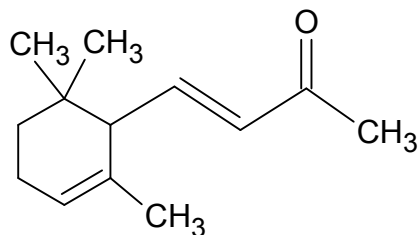
(i) making of alloys for aircrafts and racing cars

(ii) flares and fireworks

[2]

[Total: 10]

- 3 Ionones are a significant contributor to the aroma of roses. They are found in a variety of essential oils and are important chemicals in perfumery. One such ionone is the α -ionone as shown below.



- (a) Identify the functional groups present in α -ionone. [1]

- (b) State what you would see and give the structural formula of the organic products when α -ionone reacts with

(i) aqueous bromine

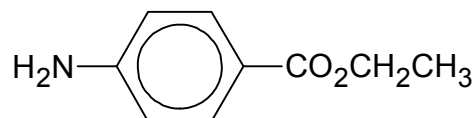
(ii) 2,4-dinitrophenylhydrazine

(iii) alkaline aqueous iodine

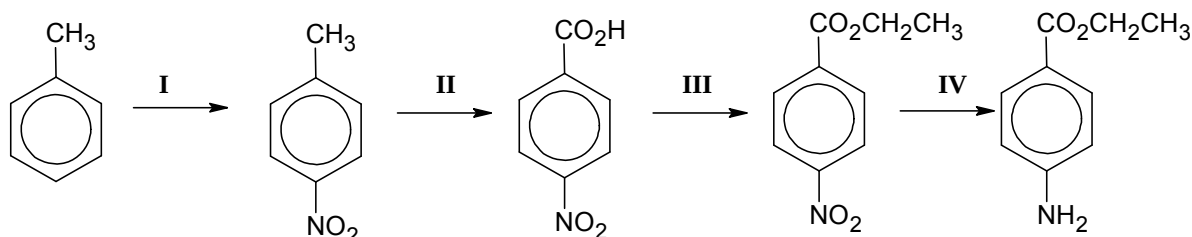
[7]

[Total: 8]

4 Benzocaine is a local anaesthetic used in sunburn ointments. Its formula is given below.



(a) Benzocaine can be made by the following reaction scheme, starting from methylbenzene.



(i) Give the reagents and conditions used for each of the stages I to III.

Stage I:

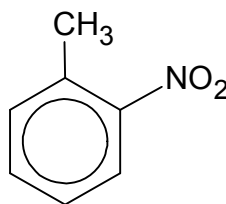
Stage II:

Stage III:

(ii) State the type of reaction undergone in stage IV.

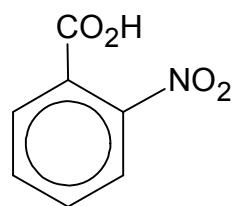
[4]

(b) (i) Besides 4-nitromethylbenzene, another isomer, 2-nitromethylbenzene is formed in stage I.



Identify the type of isomerism involved.

- (ii) When 2-nitromethylbenzene is subjected to the same set of reagents and conditions in stage **II**, 2-nitrobenzoic acid is obtained.



Which of the two carboxylic acids, 2-nitrobenzoic acid or 4-nitrobenzoic acid, would you expect to have a lower melting point? Explain your answer.

[3]

[Total: 7]

Section B (40 marks)

Answer **two** questions from this section on separate answer paper.

5 This question is about organic acids.

- (a) (i) Carboxylic acid, **A**, is the main component of vinegar with a distinctive sour taste and pungent smell. On combustion, 0.00250 mol of **A** reacted exactly with 120 cm³ of oxygen to produce 120 cm³ of carbon dioxide and 0.090 g of water at room temperature and pressure. Determine the molecular formula of **A** and hence identify it.

- (ii) Explain, with the aid of a diagram, why **A** is soluble in water.

[5]

- (b) Lactic acid, CH₃CH(OH)COOH, also known as milk acid, plays a role in a number of biochemical processes. A 0.10 mol dm⁻³ lactic acid solution has a pH of 2.43.

- (i) Show, by means of calculations, that lactic acid is a weak acid.

- (ii) Write an expression for the acid dissociation constant, K_a , of lactic acid. Hence, calculate a value of K_a of lactic acid.

- (iii) When sodium lactate is added to lactic acid, a buffer solution is obtained. Explain, with the aid of equations, how the mixture can act as a buffer on the addition of a small amount of acid and alkali.

[6]

- (c) Valeric acid or pentanoic acid, CH₃CH₂CH₂CH₂COOH, is found naturally in the perennial flowering plant, *valerian*, and has an unpleasant odour.

- (i) Give a functional group isomer of pentanoic acid.

- (ii) Pentanoic acid may be formed by heating **B** with acidified K₂Cr₂O₇. Suggest an identity of **B**, state the observation and give a balanced equation for this reaction.

- (iii) Give the structural formulae of the organic products formed when pentanoic acid reacts with:

(I) Na

(II) CH₃OH, concentrated H₂SO₄, heat

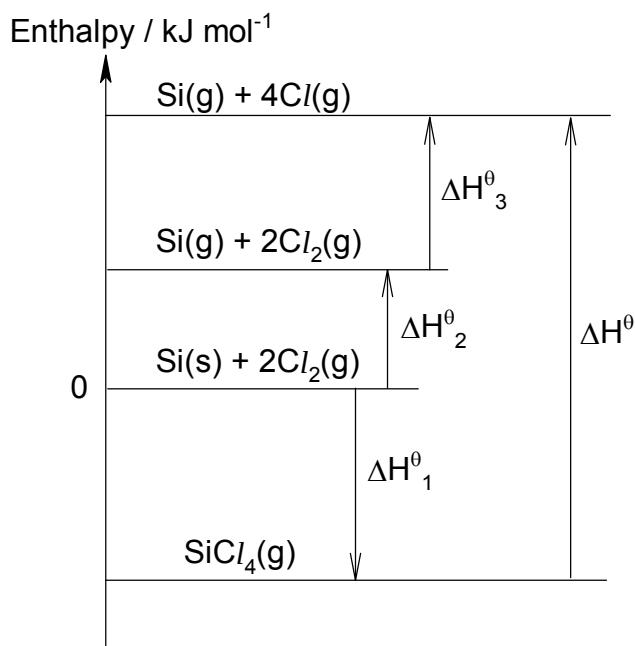
- (iv) Pentanoic acid can undergo chlorination to form chlorine-containing carboxylic acids. Suggest, with an explanation, whether 2-chloropentanoic acid or 4-chloropentanoic acid has a lower pK_a value.

[9]

[Total: 20]

6 This question is about silicon and its compounds.

- (a) (i) Give the full electronic configuration of silicon.
- (ii) Explain why the first ionisation energy of silicon is less than that of carbon. [3]
- (b) (i) Draw the dot-and-cross diagram of SiCl_4 . State the shape of the molecule and give its bond angle.
- (ii) SiCl_4 has a melting point of -70°C , but SiO_2 melts at 1700°C . Account for the difference in melting point in terms of structure and bonding.
- (iii) Write an equation for the reaction of SiCl_4 with water and state the pH of the resulting solution. [6]
- (c) The energy cycle below can be used to calculate the average Si-Cl bond energy in SiCl_4 molecule.

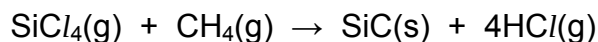


- (i) What is meant by the term *bond energy*?
- (ii) Name the standard enthalpy change represented as ΔH^θ_1 .
- (iii) Using the information provided below, calculate a value for the average Si-Cl bond energy in SiCl_4 molecule.

$$\begin{aligned}\Delta H^\theta_1 &= -610 \text{ kJ mol}^{-1} \\ \Delta H^\theta_2 &= +388 \text{ kJ mol}^{-1} \\ \Delta H^\theta_3 &= +488 \text{ kJ mol}^{-1}\end{aligned}$$

[4]

- (d) SiCl_4 reacts with CH_4 to form SiC films at 1400°C :



Using the information provided below, calculate the enthalpy change for the above reaction.

Enthalpy change of formation of $\text{SiCl}_4(\text{g}) = -687 \text{ kJ mol}^{-1}$

Enthalpy change of formation of $\text{CH}_4(\text{g}) = -74 \text{ kJ mol}^{-1}$

Enthalpy change of formation of $\text{SiC}(\text{s}) = -65 \text{ kJ mol}^{-1}$

Enthalpy change of formation of $\text{HCl}(\text{g}) = -92 \text{ kJ mol}^{-1}$

[2]

- (e) X is another element of the third period and it forms a chloride, XCl_y .

XCl_y is a liquid which has a boiling point of 76°C and fumes in air. After mixing 0.010 mol of XCl_y with water, the resulting solution required 100 cm^3 of 0.30 mol dm^{-3} silver nitrate for complete precipitation of the chloride ion.

(i) Predict the type of structure and bonding in the chloride, XCl_y .

(ii) Calculate a value of y.

(iii) Suggest, with reasons, the identity of element X.

[5]

[Total: 20]

- 7 (a) Organic compound **L** is found in plants called "Hops", which is used to treat insomnia. A molecule of **L** has the molecular formula, $C_5H_{10}O$.

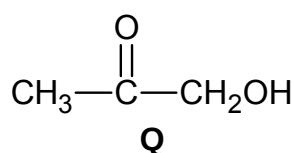
No reaction occurs between **L** and 2,4-dinitrophenylhydrazine. However, **L** produces **M**, $C_5H_{12}O$, with hydrogen with nickel catalyst.

No change is observed when **L** is warmed with acidified $K_2Cr_2O_7$.

When **L** is heated with an excess of concentrated sulfuric acid, **N**, with molecular formula C_5H_8 , is produced.

N reacts with hot concentrated acidified $KMnO_4$ to produce **P**, $C_3H_4O_3$. A colourless gas, which forms a white precipitate in limewater, is liberated.

P can also be produced by the reaction of compound **Q** with acidified $KMnO_4$ and heating.



Deduce the structures of compounds **L**, **M**, **N**, and **P**. Explain the chemistry of the reactions described.

[9]

- (b) Compound **R**, CH_3CH_2COOH , is an isomer of compound **Q** from (a).

An excess of small pieces of sodium carbonate was allowed to react with 100 cm^3 of 0.100 mol dm^{-3} of **R** at room temperature and pressure.

The volume of carbon dioxide gas liberated was measured at regular intervals and the results are tabulated below.

Time / s	5	10	15	20	25	30	35	45
Volume of CO_2 / cm^3	34	60	78	92	104	112	117	120

- Show, by means of calculations, that the maximum volume of carbon dioxide liberated was 120 cm^3 .
- Plot a graph of volume of CO_2 against time for the experiment. Label this graph as Experiment I.
- Use your graph to determine the order of reaction with respect to **R**, showing your working clearly. Hence, give the rate equation for the reaction.
- The experiment was repeated using 100 cm^3 of **R** with an initial concentration of $0.0500\text{ mol dm}^{-3}$. On the graph you have plotted for (b)(ii), sketch the graph of volume of CO_2 against time for this new experiment. Label the graph as Experiment II.

- (v)** State how the rate constant and rate of reaction will be affected when the experiment is conducted at a higher temperature.

[11]

[Total: 20]

End of Paper